

Analyzing an Organism's Sensors using Maximum Entropy Models with Bias, Variance, and Confusion Matrices

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- Organisms have sensors that communicate information about the environment.
 - Noisy replicas, memories, predictions
- Past analysis: Mutual Information
- Alternative: Stimulus-dependent Maximum Entropy models combined with bias/variance calculations or confusion matrices.

The Maximum Entropy Model

- Key Idea: Finding the most general probability distribution governing a constrained physical system
- Mathematically equivalent to minimizing Kullback-Liebler divergence
- The typical distribution follows¹

$$p(\mathbf{x}) = \frac{1}{Z} \exp(\lambda_\mu f^\mu(\mathbf{x})), \text{ where}$$

- $\mathbf{x}$: State vector
- λ_μ : Lagrange multipliers
- f^μ : Constraints

Stimulus-dependent models

$p(\mathbf{x}) \mapsto p(s \mid x)$, where s is our sensor state and x is the environmental state

¹Schneidmann et al., *Nature* 2006

Deriving an Environmental Estimator

- An application of MLE: choose $\hat{x}(s) = \arg \max_x p(x \mid s)$!
- Deriving the environmental estimator: $\hat{x}(s) = \arg \max_x p(s \mid x)p(x)/p(s)$ (via Bayes)

Quantifying MaxEnt Estimators

Ordered Data

- * Ex: estimating ligand concentrations from activity of a protein
- * Use bias for accuracy, variance for noisiness, and mean-squared error for overall quality via

$$\text{MSE} = \text{Bias}^2 + \text{Var}$$

Categorical Data

- * Ex: estimating binary activation of a stimulus neuron given the network array
- * Use confusion matrix to quantify overall accuracy

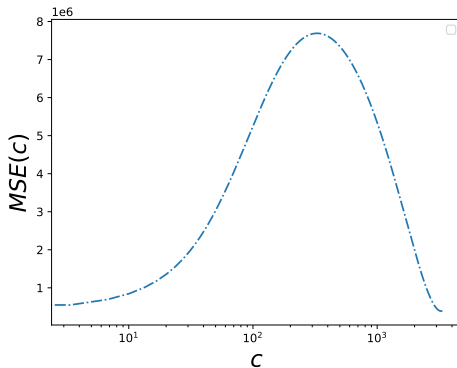
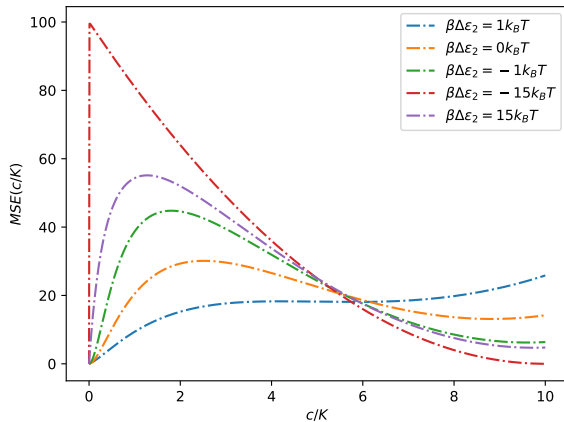


Figure: Mean-squared error for the inferred ACh ligand concentration as a function of the true ligand concentration.

Ordinal Data: Independent vs. Cooperative Sensors



- Comparing cooperativity with MSE for a protein with *two* ligands
- Second bound ligand adds interaction energy term
- Positive/negative cooperativity goodness varies with concentration

Categorical Data on Neuronal Networks: Confusion Matrices

- Basis: predict the behavior of a stimulus “neuron” x given the environmental neurons s

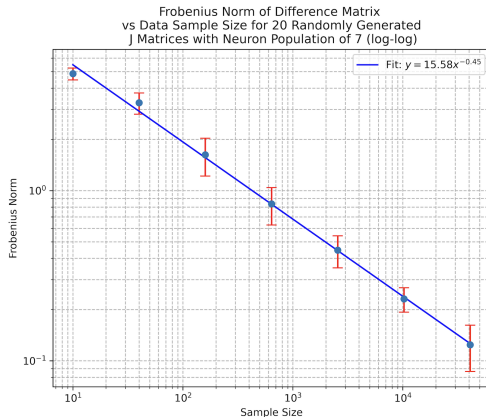
Method

- Model $\sigma = (x \ s)$
- Computationally optimize J such that $D_{\text{KL}}(p_{\text{model}} \mid p_{\text{data}})$ is minimized, where

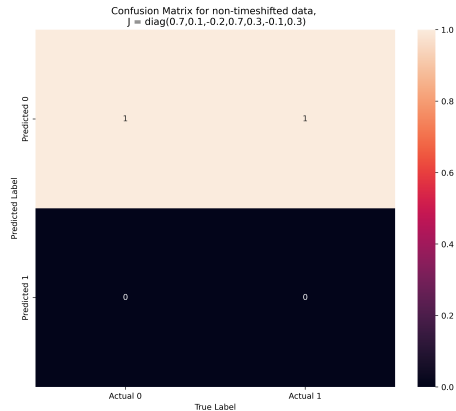
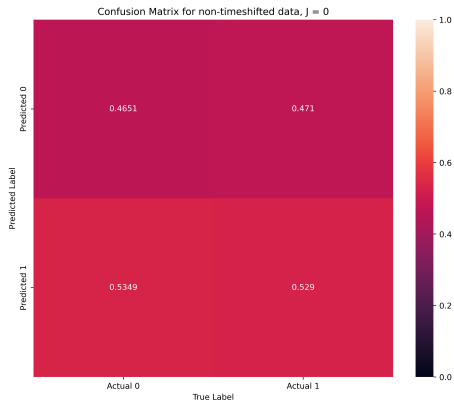
$$p_{\text{model}} = \frac{1}{Z} \exp(\sigma^{\top} J \sigma)$$

- Compute likelihood $L(s)$ via

$$L(s) = \exp(-J_{xx} - 2J_{xs}s)$$



Quality Control on our MaxEnt Model



Confusion Matrices for Local/Global Stimulation

	Actual 0	Actual 1
Predicted 0	100%	91.54%
Predicted 1	0%	8.46%

Table 1. In optogenetic/global stimulation trials, the confusion matrix for the estimator in predicting the stimulus from neural activity.

	Actual 0	Actual 1
Predicted 0	100%	26.9%
Predicted 1	0%	73.1%

Table 2. In electrical/focal stimulation trials, the confusion matrix for the estimator in predicting the stimulus from neural activity.

We'd like to thank Professor Sarah Marzen!